

Emine Berna Durmus

Protein Biochemistry and Biophysics | Assay and Method Development |
Quantitative Data Generation & Analysis

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Lausanne, Vaud



Strengths

- Hands-on with protein expression, purification, and in vitro characterization of protein function, stability, and protein-protein interactions.
- Hybrid wet-lab and computational scientist. Runs the whole chain from sample preparation and assay design through data generation to custom Python analysis with ML-based image processing.
- Moves into unfamiliar areas, builds the method, documents it, and hands it over so the lab can keep using it.

Education

Ph.D. in Biotechnology & Bioengineering , EPFL, Switzerland	2022 – 2026
M.Sc. in Molecular Biosciences , Heidelberg University, Germany	2018 – 2021
B.S. in Molecular Biology & Genetics , Istanbul Technical University, Turkiye	2015 – 2018

Professional Experience

Doctoral researcher **2022 – 2026**

Quantitative imaging of lipid droplet dynamics and mitochondria interactions in single cells

Laboratory of Experimental Biophysics, EPFL

- Started and led a new research line on lipid droplets in a lab with no prior work on the topic. Chose the questions, optimized the protocols, and produced the data showing lipid droplet mobility and mitochondrial interactions shift with metabolic state, with potential implications for understanding functional differences of lipid droplets.
- Built the experimental workflow: sample preparation, imaging protocols across expansion microscopy, iSIM, Airyscan confocal, spinning-disk confocal, and label-free microscopy, and the quantification that followed.
- Wrote custom Python image analysis pipelines from scratch, with ML-based image processing for quantification and tracking.
- Ran 2 other project lines at the same time, wrote and maintained the protocols, and co-organized international collaborations with researches from other fields.

Skills

Protein biochemistry & biophysics: Protein expression and purification (Ni-NTA affinity, anion-exchange, size-exclusion chromatography on ÄKTA), NanoDSF, light-scattering aggregation assays, SDS-PAGE, 2D-PAGE, western blot, FRET assays, liquid chromatography-tandem mass spectrometry.

Molecular biology & DNA assembly: Molecular cloning and DNA assembly (Golden Gate, Gibson assembly, PCR, restriction-ligation), CRISPR-Cas9 engineering, mammalian and bacterial cell culture, *C. elegans* work, DNA and RNA isolation, qPCR, RT-qPCR, gel electrophoresis, immunofluorescence, micropatterning, electroporation.

Data Analysis & computation: Python, ML-based image processing, multi-dimensional image and data processing pipelines, quantitative experimental design, statistical analysis, data visualization, AI-assisted coding (Claude Code, Github copilot), PyMOL.

Imaging & instrumentation: Super-resolution microscopy (STORM, iSIM, expansion microscopy), Airyscan confocal, spinning-disk confocal, live-cell imaging, label-free microscopy, polarization microscopy.

Ways of working: Method and protocol development, experimental documentation and data organization, cross-disciplinary collaboration, scientific writing, project management.

Projects

Research Scientist

June 2021 - November 2021

Troubleshooting and super-resolution imaging

Laboratory of Experimental Biophysics, EPFL

- Diagnosed and resolved an imaging artifact in super-resolution microscopy through systematic troubleshooting and applying prior biochemistry experience, which restored image quality and enabled data generation for a project on microtubules.
- Co-designed a custom metal part with the mechanical workshop to reduce sample-air exposure during imaging. Specified the requirements, iterated on the design with the engineering team, and integrated the part into the imaging workflow.

Master Thesis Researcher

July 2020 – March 2021

*Long-term live imaging of *C. elegans* and quantitative single-nucleus analysis*

EMBL, Heidelberg

- Picked up an unsolved imaging challenge in the lab, which is long-term live imaging of *C. elegans*. The approach unlocked a new project line that continued after my departure and fed into a co-authored preprint and a Best Poster Prize at an international meiosis conference.
- Generated CRISPR-Cas9 engineered transgenic *C. elegans* strains expressing fluorescent reporters. The lab kept one strain as a working resource.
- Performed spinning-disk confocal live imaging and quantitative image analysis at the single-nucleus level in Python.

Research Intern

September 2019 – February 2020

Expression, purification and in vitro biophysical characterization of a chaperone–substrate system

Heidelberg University (ZMBH) and EMBL, Heidelberg

- Independently led an *in vitro* biophysical characterization of the molecular chaperone DnaK and its substrate RepE to investigate how chaperones recognize folded protein substrates, which is a fundamental open question in protein homeostasis.
- Expressed and purified both proteins and found working buffer and expression conditions for the substrate by systematic screening. Wrote up the protocols and results as a structured handover for the lab.
- Diagnosed and troubleshooted multiple experimental setbacks independently, characterized an unexpected protein dimerization that explained why planned experiments could not proceed, resolved a separate co-aggregation problem, and validated assay reproducibility through orthogonal cross-checks.

Research Intern

March 2019 – April 2019

In vitro reconstitution and domain-level dissection of Munc13-1 function in vesicle fusion

Heidelberg University (BZH)

- Reconstituted an *in vitro* vesicle fusion system and ran FRET-based lipid mixing assay to test how different domains of Munc13-1 protein contribute to fusion, with NanoDSF stability measurement on point mutants alongside the functional FRET readout.

Languages

English Professional | German A2 | Turkish Native

Publications

Contributed to 3 articles in high ranked journals and one co-authored preprint.

Full list on ORCID: <https://orcid.org/0009-0005-3034-4044>